

Golden Rule -> clean, well-structured text in = good chunks and good retrieval out

1. Why Preprocessing Matters

A RAG system can only retrieve what it was given cleanly. Messy text — broken encoding, stray headers/footers, mangled tables — produces messy chunks, which produces messy embeddings, which produces bad retrieval. Loading is the first and most underrated step in the pipeline.

2. Common Document Loaders

Loader	Use For	Notes
TextLoader	Plain .txt files	Simplest case; watch file encoding (use utf-8)
PyPDFLoader	PDF documents	Extracts text page-by-page; scanned/image PDFs need OCR first
DirectoryLoader	A whole folder of mixed files	Pair it with a loader_cls per file extension
CSVLoader / UnstructuredLoader	Structured or semi-structured data	Useful when you need columns/fields preserved as metadata

3. Preprocessing Steps

- Strip boilerplate: running headers, footers, repeated page numbers.
- Normalize whitespace and text encoding (aim for consistent utf-8).
- Preserve useful structure — headings, section numbers — as metadata rather than deleting it; it helps later with chunking and citations.
- Flag documents with tables/images: raw text extraction from these is often incomplete or garbled and may need a specialised parser.

4. Code Example

```
$ from langchain_community.document_loaders import PyPDFLoader

$ loader = PyPDFLoader("policy.pdf")
$ docs = loader.load()

$ print(docs[0].page_content[:300])
$ print(docs[0].metadata) # {'source': 'policy.pdf', 'page': 0}
```

5. Keeping Metadata Alive

Every loaded document should carry at least its source filename and page number in its metadata. This travels through chunking and embedding, and is what lets the final RAG answer say “according to page 4 of policy.pdf” instead of just guessing.

Pitfall

Scanned PDFs (image-only pages) have no extractable text layer — PyPDFLoader will silently return empty or near-empty strings for them. You need an OCR step (e.g. unstructured + tesseract) before these pages are usable.

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Remember

Keep metadata (source filename, page number, section) attached to every document from the very first load — you'll need it later to cite sources in the final answer.

INTERVIEW PREP

Q: What's the difference between loading a .txt file and a scanned PDF?

A: A .txt file already contains extractable text; a scanned PDF is essentially an image, so a text loader like PyPDFLoader returns nothing useful from it without an OCR step first.

Q: Why keep metadata like source and page number during loading?

A: So the final RAG answer can cite exactly where information came from, and so retrieval and debugging can trace any chunk back to its original document and location.

Q: What preprocessing mistake most commonly hurts retrieval quality?

A: Either stripping away too much structure (headings, sections) or leaving in noise (repeated headers/footers, page numbers) — both distort chunk boundaries and embedding quality.

Q: How would you handle a folder containing both PDF and text files?

A: Use a DirectoryLoader configured with a per-extension loader_cls (e.g. PyPDFLoader for .pdf, TextLoader for .txt) so each file type is parsed with the right tool.